STEEL BEAM ESTIMATION

Current Performance & Vision Impact · Second to Sixth Sets

Deterministic model = production authority. Claude Vision = shadow / research only. First Set is excluded from all aggregate KPIs.

CURRENT DETERMINISTIC MODEL — SECOND TO SIXTH SETS

Beam identification	Bar identification	Correct of detected bars	Diameter identification
85.12%	50.37%	33.95%	75.67%

Bar matching (vs GT)	Steel accuracy (kg-pooled)	Overall accuracy	Vision (Fifth Set only)
17.10%	73.59%	60.76%	+3.02 pp shadow

The model identifies 515 of 605 ground-truth beams (85.12%) and 2,100 of 4,169 ground-truth bar lines (50.37%). Of the 2,100 bars it does identify, 713 are a full MATCH on role, diameter and quantity (33.95%). Diameter is correct on 75.67% of those detected bars (WRONG_DIAMETER excluded). Combined steel is 109,580 kg versus 148,911 kg estimator (73.59% QA.2A steel accuracy on pooled kg). Overall accuracy is the established four-metric mean: beam identification, bar identification, correct identification of detected bars, and pooled steel accuracy.

These figures are the current production-authority result. They do not include Claude Vision.

1. Drawing-wise deterministic performance

Drawing set	Beam ID	Bar ID	Correct of detected	Diameter ID	Bar matching	Steel	Overall
Second Set (known)	95.52%	71.76%	35.74%	77.98%	25.65%	85.50%	72.13%
Third Set (known)	90.48%	63.52%	17.91%	68.24%	11.37%	87.89%	64.95%
Fourth Set (unseen)	78.32%	39.89%	37.33%	73.07%	14.89%	58.39%	53.48%
Fifth Set (unseen)	76.47%	40.25%	34.30%	74.23%	13.80%	61.69%	53.18%
Sixth Set (unseen)	95.86%	61.45%	38.87%	81.63%	23.89%	96.54%	73.18%
ALL FIVE SETS	**85.12%**	**50.37%**	**33.95%**	**75.67%**	**17.10%**	**73.59%**	**60.76%**

Beam ID = detected beams / GT beams. Bar ID = detected bar lines / GT bar lines. Correct of detected = MATCH / detected (QA.2A bar_accuracy). Bar matching = MATCH / GT bars. Diameter ID = (detected − WRONG_DIAMETER) / detected. Steel = QA.2A 100 − |model−estimator| / estimator; five-set row is kg-pooled, not an average of %. Overall = mean of Beam ID, Bar ID, Correct of detected, and Steel. Known = Second–Third (development). Unseen = Fourth–Sixth (generalization).

2. What the five questions mean in numbers

Question	KPI	Five-set result	Raw counts
What % of beams does the model identify?	Beam identification	85.12%	515 / 605 GT beams
What % of bars does the model identify?	Bar identification	50.37%	2,100 / 4,169 GT bar lines
Of bars identified, what % are correct?	Correct of detected bars	33.95%	713 MATCH / 2,100 detected
How accurately are diameters identified?	Diameter identification	75.67%	(2,100 − 511 WRONG_DIA) / 2,100
How close is steel quantity to the estimator?	Steel accuracy	73.59%	109,580 / 148,911 kg

3. Known (Second–Third) vs unseen (Fourth–Sixth)

Metric	Second–Third known	Fourth–Sixth unseen	All five sets
Beam identification	93.08% (121/130)	82.95% (394/475)	85.12%
Bar identification	67.25% (573/852)	46.04% (1,527/3,317)	50.37%
Correct of detected bars	26.53% (152/573)	36.74% (561/1,527)	33.95%
Diameter identification	73.00%	76.69%	75.67%
Bar matching vs GT	17.84% (152/852)	16.91% (561/3,317)	17.10%
Steel accuracy (QA.2A set-mean)	86.70%	72.21%	78.00%*
Steel accuracy (kg-pooled)	98.37%†	68.23%	73.59%
Overall (pooled steel in five-set)	68.39%‡	59.49%‡	60.76%

* QA.2A dashboard steel for multiple sets is the unweighted mean of per-set steel %. † Known kg-pooled steel is inflated by Second-Set over-estimate cancelling Third-Set under-estimate — do not read 98% as known-set quality. ‡ Group overall uses QA.2A set-mean steel. Five-set overall uses kg-pooled steel.

4. Diameter identification (detected bar lines)

QA.2A's published field diameter_accuracy_pct is an alias of bar matching accuracy (MATCH / detected) and is not used here. Diameter identification below is counted from Bar Matching rows: a detected bar is diameter-correct unless status is WRONG_DIAMETER. GT diameter is the estimator line diameter. Ø32 has only 8 detected lines — percentage unstable.

Diameter	GT bar lines	Detected	MATCH	WRONG_DIA	Diameter ID	Note
Ø8	358	159	37	3	98.11%	Usually right when found; many missing
Ø10	714	235	72	10	95.74%	Same pattern — detection is the gap
Ø12	573	224	64	31	86.16%	
Ø16	588	404	155	140	65.35%	Frequent diameter swaps
Ø20	784	431	151	186	56.84%	Weakest major diameter
Ø25	1,097	639	231	136	78.72%	Includes many spacer lines
Ø32	26	8	3	5	37.50%	Low volume — unstable
TOTAL scored Ø8–Ø32	**4,140**	**2,100**	**713**	**511**	**75.67%**	**Five-set headline**

5. Diameter-wise steel quantity — not the same as diameter identification

Quantity ratio = automated kg / estimated kg × 100. It is not accuracy. A ratio above 100% is an overestimate. Ø16 is 143% of estimator kg while Ø10 is 50%.

Diameter	Estimated kg	Automated kg	Difference kg	Abs % diff	Quantity ratio
Ø8	6,515	4,389	−2,126	32.6%	67%
Ø10	24,555	12,236	−12,319	50.2%	50%
Ø12	22,253	11,670	−10,582	47.6%	52%
Ø16	11,598	16,564	+4,965	42.8%	143%
Ø20	35,888	29,320	−6,568	18.3%	82%
Ø25	45,294	32,965	−12,328	27.2%	73%
Ø32	2,808	2,436	−372	13.2%	87%
TOTAL	**148,911**	**109,580**	**−39,330**	**26.4%**	**74%**

6. Error profile — Second to Sixth Sets

Error category	Second	Third	Fourth	Fifth	Sixth	Five-set
Missing bars	109	170	565	870	355	2,069
Wrong quantity	81	114	103	166	176	640
Wrong diameter	61	94	101	151	104	511
Extra bars	74	94	51	123	81	423
Wrong role	30	23	20	45	40	158
Beam missing	3	6	31	44	6	90
Beam mismatch	0	4	6	0	4	14
Steel difference	1	1	1	1	1	5
Total classified errors	**359**	**506**	**878**	**1,400**	**767**	**3,910**

Missing bars dominate unseen sets. Diameter and quantity errors remain material even when a bar is detected. This is why 50% bar identification does not become 50% correct steel.

7. Claude Vision — experimental, Fifth Set only

Engineering-level Vision-assisted recompute has currently been demonstrated on the Fifth Set. No valid engineering recompute exists for Second, Third, Fourth or Sixth Sets. A five-set Vision KPI is therefore not calculated. P2.5.8 is the authoritative Vision engineering result. P2.5.9–P2.5.11 are later safety/arbitration experiments and are not production Vision accuracy.

Fifth Set steel	Deterministic (production)	Vision-assisted (P2.5.8 shadow)	Estimator
Steel kg	36,271.794	38,049.747	58,796.332
Steel accuracy	61.69%	64.71%	—
Absolute error	38.31%	35.29%	—

Accuracy lift	Error reduction	Stirrup accuracy	Beams ↑ / = / ↓
+3.02 pp	**7.88%**	**35.44% → 44.35%**	**21 / 150 / 14**

P2.5.8 replayed frozen P2.5.7 Claude results (0 new API calls) onto a sandbox copy of the reinforcement model. Practical value was OCR-difficult stirrups and slash spacing such as 100/125/100 that SI.1 truncated to the first token. Promoted fields: diameter 29, legs 29, spacing 35, role 0. Production mutation = 0; steel / BBS / Excel production difference = 0.

P2.5.8 DECISION · NEGATIVE · NOT PRODUCTION-SAFE

Do not present 64.71% as current production accuracy. Fourteen beams worsened. Unrestricted new-stirrup inserts were rejected. P2.5.9–P2.5.11 gate those inserts (UNKNOWN-only, then evidence HOLD) and drop the lift to about +0.3 pp with zero worsenings.

8. Current workflow value — model-assisted, not autonomous

DXF → beam discovery → reinforcement interpretation → stirrup interpretation → steel calculation → BBS → Excel output → estimator verification.

Already automated: beam discovery (~85% of GT beams), annotation parsing, stirrup parsing, IS 456 steel calculation, BBS and Excel. Still required: finding the ~15% missed beams, the ~50% missed bars, and correcting ~66% of detected bars that are not a full MATCH. Positioning: model-assisted estimation / estimator first-pass support.

9. Indicative estimator time-saving potential

No estimator time-and-motion study, manual-vs-model hours, or productivity log exists in the project. Accuracy percentages are not time saved. 50% bar identification is not 50% time saved, because missed bars and incorrect detected bars still require drawing reading.

If used today as a first-pass worksheet	Indicative range	Basis (not a measurement)
Beam take-off / marking	Partial save	85% of GT beams already listed
Bar schedule compilation	Limited save	Half of GT bars missing; 2 in 3 detected bars need edit
Steel calc / BBS / Excel typing	High save	Fully automated for whatever the model produced
Verification / red-line	Still required	Unseen-set steel 58–62% except Sixth Set
Overall estimation time	20–35%	Indicative only — requires time-and-motion study

Indicative time-saving potential today: 20–35% of manual estimation time, if the Excel is used as a first pass and the estimator still verifies every beam. This is not reliably measurable until a timed estimator study is run. Computer pipeline runtime (e.g. Fifth Set ~2.8 h) is model processing, not estimator saving.

10. Current status

Item	Value
Beam identification	85.12%
Bar identification	50.37%
Correct identification of detected bars	33.95%
Diameter identification accuracy	75.67%
Bar matching accuracy (MATCH / GT)	17.10%
Deterministic steel accuracy (kg-pooled)	73.59%
Overall accuracy	60.76%
Vision-assisted engineering improvement	Fifth Set only: 61.69% → 64.71% (+3.02 pp, shadow)
Indicative estimator time saving	20–35% · not yet reliably measurable
Current usage	Model-assisted estimation / estimator first-pass support
Production authority	Deterministic model
Claude Vision	Controlled shadow / research · P2.5.8 NEGATIVE for unrestricted promotion

Claude Vision has demonstrated a real engineering lift on difficult stirrup/OCR cases, but unrestricted promotion worsened 14 Fifth-Set beams. P2.5.9–P2.5.11 are making recovery engineering-safe rather than maximising aggregate accuracy. Production readiness is not claimed.

11. Methodology, sources, limitations

Second–Third: Version 9 QA.2B.1 GroundTruth_Benchmark_Report.xlsx (known development sets). Fourth–Sixth: Version 10 QA.3.0 Generalization_Benchmark_Report.xlsx and per-set benchmark_result.json (unseen production runs qa2_*_20260806). Counts are Excel Bar Detection / Bar Matching integers, not rounded percentages from the old six-set PDF. Steel kg from QA.2A metric8. Diameter kg from Diameter Steel Quantity sheets. Vision from P2.5.8_STATUS.md. First Set excluded (GT 18 beams / 85 bars / 1,424 kg). Limitations: no estimator time study; diameter ID is status-based (PARTIAL_MATCH residual exists); Ø1/Ø2 rows in quantity sheets are parser artefacts and were omitted; Vision engineering recompute exists only for Fifth Set.